

PCA and Latent Factor Models (II)

MFE5340: AI in FE: Quantitative Investment

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Spring 2026



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Table of Contents

1 Recap

- Recap
- Risk Premium PCA
- PCA with Economic Targets
- Instrumented PCA
- A New Norm Beyond PCA



The Players

- $\mathbf{R}$ ($T \times N$ matrix): excess returns of assets (T periods and N assets).
- $\mathbf{F}$ ($T \times K$ matrix): excess returns of factors (where $K \leq N$).
- $\mathbf{\Lambda}$ ($N \times K$ matrix): assets' factor loadings.
- $\tilde{\mathbf{R}}$ ($T \times N$ matrix): column-wise **demeaned** version of $\mathbf{R}$,
- $\tilde{\mathbf{F}}$ ($T \times K$ matrix): column-wise **demeaned** version of $\mathbf{F}$.
- $\bar{\boldsymbol{R}}$ ($N \times 1$ vector): the sample mean excess returns of N assets.
- $\bar{\boldsymbol{F}}$ ($K \times 1$ vector): the sample average excess returns of K factors.

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- Running PCA to the covariance matrix $\mathbf{S} = \frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}} = \frac{1}{T}\mathbf{R}^\top\mathbf{R} - \bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top$ is equivalent to solving the following least squares optimization problem:

$$\min_{\boldsymbol{\Lambda},\tilde{\mathbf{F}}} \frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T}\left(\tilde{R}_{tn} - \tilde{\boldsymbol{F}}_t^\top\boldsymbol{\Lambda}_n\right)^2 \quad \text{with } \tilde{\mathbf{F}} = \tilde{\mathbf{R}}\boldsymbol{\Lambda}(\boldsymbol{\Lambda}^\top\boldsymbol{\Lambda})^{-1},$$

  where $\tilde{\boldsymbol{F}}_t^\top$ is the tth row of $\tilde{\mathbf{F}}$ (and hence $\tilde{\boldsymbol{F}}_t^\top$ is a $1 \times K$ row vector) and $\boldsymbol{\Lambda}_n^\top$ is the nth row of $\boldsymbol{\Lambda}$ (and hence $\boldsymbol{\Lambda}_n^\top$ is a $1 \times K$ row vector).

- In matrix form:

$$\min_{\boldsymbol{\Lambda},\tilde{\mathbf{F}}} \frac{1}{NT}\|\tilde{\mathbf{R}} - \tilde{\mathbf{F}}\boldsymbol{\Lambda}^\top\|_F^2.$$

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Least Square Perspective

- With the restriction $\tilde{\mathbf{F}} = \tilde{\mathbf{R}}\mathbf{\Lambda}(\mathbf{\Lambda}^\top\mathbf{\Lambda})^{-1}$, we can write:

$$\tilde{\mathbf{R}} - \tilde{\mathbf{F}}\mathbf{\Lambda}^\top = \tilde{\mathbf{R}} - \tilde{\mathbf{R}}\mathbf{\Lambda}(\mathbf{\Lambda}^\top\mathbf{\Lambda})^{-1}\mathbf{\Lambda}^\top.$$

- Complement projection matrix:

$$\mathbf{M} = \mathbf{I}_N - \mathbf{\Lambda}(\mathbf{\Lambda}^\top\mathbf{\Lambda})^{-1}\mathbf{\Lambda}^\top.$$

- Then:

$$\min_{\mathbf{\Lambda},\tilde{\mathbf{F}}} \frac{1}{NT}\|\tilde{\mathbf{R}}\mathbf{M}\|_F^2.$$

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- Trace operation (tr()): $\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}^\top \mathbf{A}) = \text{tr}(\mathbf{A}\mathbf{A}^\top)$.
- Therefore:
$$\frac{1}{NT}\|\tilde{\mathbf{R}}\mathbf{M}\|_F^2 = \frac{1}{NT}\text{tr}\left[(\tilde{\mathbf{R}}\mathbf{M})^\top(\tilde{\mathbf{R}}\mathbf{M})\right] = \frac{1}{NT}\text{tr}\left[\mathbf{M}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}\mathbf{M}\right].$$
- This equation can be understood from right to left: minimizing $\frac{1}{NT}\text{tr}\left[\mathbf{M}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}\mathbf{M}\right]$ is equivalent to minimizing $\frac{1}{NT}\|\tilde{\mathbf{R}}\mathbf{M}\|_F^2$. And $\frac{1}{NT}\|\tilde{\mathbf{R}}\mathbf{M}\|_F^2$ is merely the matrix form of the original objective function, where solving the original objective function is equivalent to performing PCA on the sample covariance matrix $\mathbf{S} = \frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$.

The middle matrix $\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ in $\frac{1}{NT}\text{tr}\left[\mathbf{M}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}\mathbf{M}\right]$ is nothing but a scaled version of $\mathbf{S}$.

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- Traditional latent factor research applies PCA to $\mathbf{S} = \frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$, which is equivalent to solving:

$$\min_{\mathbf{\Lambda},\tilde{\mathbf{F}}} \frac{1}{NT}\|\tilde{\mathbf{R}} - \tilde{\mathbf{F}}\mathbf{\Lambda}^\top\|_F^2.$$

- However, the above equivalence tells us that we can **adjust the objective function** of the optimization problem, and once it is transformed into the form of minimizing $\frac{1}{NT}\text{tr}\left[\mathbf{MAM}\right]$ through the trace operation, solving this adjusted optimization problem is equivalent to performing PCA on (a scaled version of) $\mathbf{A}$ (e.g. $\frac{1}{T}\mathbf{A}$).

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Table of Contents

2 Risk Premium PCA

- Recap
- Risk Premium PCA
- PCA with Economic Targets
- Instrumented PCA
- A New Norm Beyond PCA

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- Recall that covariance matrix does not contain first-moment information of returns.
- Consider the following new optimization problem that minimizes a weighted sum of both time-series variation of returns and the cross-sectional pricing errors:

$$\min_{\mathbf{\Lambda},\mathbf{F}} \frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T}\left(R_{tn}-\boldsymbol{F}_t^\top\mathbf{\Lambda}_n\right)^2+\frac{\gamma}{N}\sum_{n=1}^{N}\left(\bar{R}_n-\mathbf{\Lambda}_n^\top\bar{\boldsymbol{F}}\right)^2 \quad \text{with } \mathbf{F}=\mathbf{R}\mathbf{\Lambda}(\mathbf{\Lambda}^\top\mathbf{\Lambda})^{-1},$$

  where γ is an coefficient that determines the relative strengths of the two components.
- In this new objective function, we consider not only the time-series variation of returns (1st term), but also the cross-sectional pricing errors (2nd term).

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- Rewrite the objective function in matrix form (check this by yourself!!!):

$$\frac{1}{NT}\|\mathbf{R}-\mathbf{F}\boldsymbol{\Lambda}^\top\|_F^2+\frac{\gamma}{N}\|\bar{\boldsymbol{R}}-\boldsymbol{\Lambda}\bar{\boldsymbol{F}}\|_F^2.$$

- We can show that, using trace operation, the objective function can expressed as:

$$\frac{1}{NT}\text{tr}\left[\mathbf{M}\left(\mathbf{R}^\top\mathbf{R}+\gamma T\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top\right)\mathbf{M}\right].$$

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$$\frac{1}{NT}\text{tr}\left[\mathbf{M}\left(\mathbf{R}^\top\mathbf{R}+\gamma T\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top\right)\mathbf{M}\right].$$

- It is immediately clear that the middle matrix is $\mathbf{R}^\top\mathbf{R}+\gamma T\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top$, and that solving the new optimization problem is equivalent to conducting eigen-decomposition to the following matrix:

$$\boldsymbol{\Sigma}_{\text{RP}}=\frac{1}{T}\left[\mathbf{R}^\top\mathbf{R}+\gamma T\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top\right]=\frac{1}{T}\mathbf{R}^\top\mathbf{R}+\gamma\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top.$$

- This is called the **Risk Premium PCA** (RP-PCA) since the objective function considers both time-series and cross-sectional information of asset returns.
- When $\gamma=-1$, the $\boldsymbol{\Sigma}_{\text{RP}}$ reduces to the sample covariance matrix, and RP-PCA coincides with traditional PCA.

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Table of Contents

3 PCA with Economic Targets

- Recap
- Risk Premium PCA
- PCA with Economic Targets
- Instrumented PCA
- A New Norm Beyond PCA

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- Bryzgalova et al. (2023) proposed a general PCA framework that allows users to incorporate cross-sectional, time-series, or both information into the objective function. (I will talk about how to integrate cross-sectional information into traditional PCA in detail.)

- Portfolios constructed by sorting based on firm characteristics exhibit monotonic returns. Suppose we consider L such portfolios and let the $N \times L$ matrix $\mathbf{W}^{XS}$ represent the weights of stocks within these portfolios. Returns of these portfolios can be expressed as $\mathbf{R}^{XS} = \mathbf{R}\mathbf{W}^{XS}$, where $\mathbf{R}^{XS}$ is a $T \times L$ matrix.

- Each of the L portfolios is merely a linear combination of the N stocks. Therefore, their factor exposures are also a weighted average of the individual stock factor exposures:

$$\underbrace{\mathbf{\Lambda}^{XS}}_{L\times K} = \underbrace{(\mathbf{W}^{XS})^{\top}}_{L\times N}\underbrace{(\mathbf{\Lambda})}_{N\times K}.$$

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- We want the factor exposures of these portfolios to the latent factors, $\mathbf{\Lambda}^{XS}$, and their returns, $\mathbf{R}^{XS}$, to align with each other. (Why?)

- This alignment will guide the estimation process of the latent factors, deviating from the traditional PCA by incorporating the cross-sectional information contained in these L portfolios.

- To this end, consider the following least squares problem:

$$\min_{\mathbf{\Lambda},\mathbf{F}} \frac{1}{NT} \sum_{n=1}^{N} \sum_{t=1}^{T} \left(R_{tn} - \mathbf{F}_t^\top \mathbf{\Lambda}_n \right)^2 + \frac{\gamma}{LT} \sum_{l=1}^{L} \sum_{t=1}^{T} \left(R_{tl}^{XS} - \mathbf{F}_t^\top (\mathbf{\Lambda}_l^{XS}) \right)^2,$$

where $(\mathbf{\Lambda}_l^{XS})^\top$ is the lth row of $\mathbf{\Lambda}^{XS}$, and γ is an coefficient that determines the relative strengths of the two components. **The second term represents the new cross-sectional information.**

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- In matrix form, the second term becomes:

$$\frac{\gamma}{LT}\sum_{l=1}^{L}\sum_{t=1}^{T}\left(R_{tl}^{XS}-\mathbf{F}_t^{\top}(\mathbf{\Lambda}_l^{XS})\right)^2=\frac{\gamma}{LT}\|\mathbf{R}\mathbf{W}^{XS}-\mathbf{F}\mathbf{\Lambda}^{\top}\mathbf{W}^{XS}\|_F^2.$$

- We do not impose constraint that L portfolios are independent. Therefore, they may contain some redundant information. To address this, introduce the $N\times N$ projection matrix $\mathbf{P}$:

$$\mathbf{P}=\mathbf{W}^{XS}\left((\mathbf{W}^{XS})^{\top}(\mathbf{W}^{XS})\right)^{-1}(\mathbf{W}^{XS})^{\top}.$$

- Transform the second term using $\mathbf{P}$:

$$\frac{\gamma}{LT}\|\mathbf{R}\mathbf{P}-\mathbf{F}\mathbf{\Lambda}^{\top}\mathbf{P}\|_F^2.$$

- The new second term can be understood as projecting the returns of the original N stocks onto the column space of $\mathbf{W}^{XS}$ and minimizing the pricing error of these L portfolios after the projection.

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- The objective function in matrix form:

$$\frac{1}{NT}\|\mathbf{R}-\mathbf{F}\boldsymbol{\Lambda}^{\top}\|_F^2+\frac{\gamma}{LT}\|\mathbf{RP}-\mathbf{F}\boldsymbol{\Lambda}^{\top}\mathbf{P}\|_F^2.$$

- To solve this least squares, I aim to derive from this matrix-form objective function an equivalent matrix for which we can perform PCA. To do so, I will introduce a more general and powerful approach.

- I am indebt to Prof. WANG, Xi of Peking University (School of Economics) for his insightful discussion with me regarding this approach!

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- Rewirte the objective function as:

$$\frac{1}{NT}\left\{\|\mathbf{R}-\mathbf{F}\boldsymbol{\Lambda}^\top\|_F^2+\frac{\gamma N}{L}\|\mathbf{RP}-\mathbf{F}\boldsymbol{\Lambda}^\top\mathbf{P}\|_F^2\right\}.$$

- It can be shown that (after long but elegant derivation!) that, to solve this least squares, we can apply eigen-decomposition to the following $T\times T$ matrix:

$$\mathbf{R}\left(\mathbf{I}_N+\frac{\gamma N}{L}\mathbf{P}\right)\mathbf{R}^\top.$$

- Once we back out the first K eigenvectors from eigen-decomposition, they will serve as the **factor returns** of first K factors, and we can derive factor exposures as:

$$\hat{\boldsymbol{\Lambda}}=\mathbf{R}^\top\hat{\mathbf{F}}.$$

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Table of Contents

4 Instrumented PCA

- Recap
- Risk Premium PCA
- PCA with Economic Targets
- Instrumented PCA
- A New Norm Beyond PCA

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- PCA, RP-PCA, or PCA with economic targets, are essentially **static models**. In these models, the estimated factor exposures $\mathbf{\Lambda}$ do not vary over time.

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- PCA, RP-PCA, or PCA with economic targets, are essentially **static models**. In these models, the estimated factor exposures $\mathbf{\Lambda}$ do not vary over time.
- However, characteristics of companies change over time. For instance, a small-cap company may develop into a large-cap company over time. **This means that a company's stock exposure to factors is likely to be time-varying rather than static.**

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- Another shortcoming of traditational methods (not including PCA with economic targets) is that they mainly rely on the information contained in returns and do not explicitly incorporate other information related to future stock returns, such as firm characteristics.

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Motivation

4 Instrumented PCA

- PCA, RP-PCA, or PCA with economic targets, are essentially **static models**. In these models, the estimated factor exposures $\mathbf{\Lambda}$ do not vary over time.
- However, characteristics of companies change over time. For instance, a small-cap company may develop into a large-cap company over time. **This means that a company's stock exposure to factors is likely to be time-varying rather than static.**
- Another shortcoming of traditational methods (not including PCA with economic targets) is that they mainly rely on the information contained in returns and do not explicitly incorporate other information related to future stock returns, such as firm characteristics.
- A natural extension is to consider **dynamic** factor models, where stock factor exposures change over time, and incorporate information other than returns into the model. A representative work is the Instrumented PCA (IPCA) proposed by Kelly, Pruitt and Su (2019).

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- For time t, consider the following latent factor model:

$$\underbrace{\boldsymbol{r}_{t+1}}_{N\times 1} = \underbrace{\boldsymbol{\Lambda}_t}_{N\times K}\underbrace{\boldsymbol{f}_{t+1}}_{K\times 1} + \underbrace{\boldsymbol{e}_{t+1}}_{N\times 1}.$$

- Consider **conditional** factor loadings as functions of L observable firm characteristics (known as *instruments*):

$$\underbrace{\boldsymbol{\Lambda}_t}_{N\times K} = \underbrace{\mathbf{Z}_t}_{N\times L}\underbrace{\boldsymbol{\Gamma}}_{L\times K},$$

  where $\mathbf{Z}_t$ stores the firm characteristics data of time t, and $\boldsymbol{\Gamma}$ maps $\mathbf{Z}_t$ to $\boldsymbol{\Lambda}_t$; $\boldsymbol{\Gamma}$ needs to be estimated.

- Combining them together gives:

$$\boldsymbol{r}_{t+1} = \mathbf{Z}_t\boldsymbol{\Gamma}\boldsymbol{f}_{t+1} + \boldsymbol{e}_{t+1}.$$

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$$\boldsymbol{r}_{t+1} = \mathbf{Z}_t \boldsymbol{\Gamma} \boldsymbol{f}_{t+1} + \boldsymbol{e}_{t+1}.$$

- $\boldsymbol{\Gamma}$ does not have a subscript t, implying that the model assumes it does not change over time. In other words, all variations in individual stock factor exposures are captured through the time-varying firm characteristics $\mathbf{Z}_t$.
- $\boldsymbol{\Gamma}$ is crucial as it allows the model to choose a reduced-rank combination of the L candidate characteristics that best describe the factor loading structure.
- The goal is to find $\boldsymbol{\Gamma}$ and $\mathbf{F}$ that minimize the following error:

$$\min_{\boldsymbol{\Gamma},\mathbf{F}} \sum_{t=1}^{T-1} \left(\boldsymbol{r}_{t+1} - \mathbf{Z}_t \boldsymbol{\Gamma} \boldsymbol{f}_{t+1}\right)^\top \left(\boldsymbol{r}_{t+1} - \mathbf{Z}_t \boldsymbol{\Gamma} \boldsymbol{f}_{t+1}\right).$$

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- The estimation is similar to traditional PCA, i.e., we can optimize the least squares to find the latent factors $\{\boldsymbol{f}_t\}$ and loadings $\boldsymbol{\Gamma}$, by using alternating least squares.
- Given estimate of $\boldsymbol{\Gamma}$, we have:

$$\boldsymbol{f}_{t+1} = \left(\boldsymbol{\Gamma}^\top \mathbf{Z}_t^\top \mathbf{Z}_t \boldsymbol{\Gamma}\right)^{-1} \boldsymbol{\Gamma}^\top \mathbf{Z}_t^\top \boldsymbol{r}_{t+1}.$$

- Given estimates of the series $\{\mathbf{f}_t\}, t = 1, \cdots, T-1$, we have

$$\text{vec}(\boldsymbol{\Gamma}^\top) = \left(\sum_{t=1}^{T-1} \mathbf{Z}_t^\top \mathbf{Z}_t \otimes \boldsymbol{f}_{t+1} \boldsymbol{f}_{t+1}^\top\right)^{-1} \left(\sum_{t=1}^{T-1} \left[\mathbf{Z}_t \otimes \boldsymbol{f}_{t+1}^\top\right]^\top \boldsymbol{r}_{t+1}\right).$$

  where $\otimes$ is Kronecker product.

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- If $\mathbf{A}$ is an $m \times n$ matrix and $\mathbf{B}$ is a $p \times q$ matrix, then the Kronecker product $\mathbf{A} \otimes \mathbf{B}$ is the $pm \times qn$ block matrix:

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ \vdots & \ddots & \vdots \\ a_{m1}\mathbf{B} & \cdots & a_{mn}\mathbf{B} \end{bmatrix}.$$

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- Let's go through the derivation togather.
- Lecture note has all the details.
- **This part of the content is not required for assignments or exams. It is more about familiarizing yourselves with the process, as it may be helpful in your future studies and work.**

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- In his talk at the 2018 Utah Winter Finance Conference, Bryan Kelly offered a managed portfolio interpretation for the IPCA model, which I find very intuitive.
- The IPCA estimator is approximately equivalent to:
  - For each characteristic $l = 1, \cdots, L$ in each period t, run a cross-section regression of $R_{i,t}$ on $Z_{i,l,t-1}$, estimates $X_{l,t}$, i.e., return of the managed portfolio associated with characteristic l.
  - From $T \times L$ matrix $\mathbf{X}$ of characteristic-managed portfolio returns, IPCA factor estimates $\approx$ first K PCs.
- According to this interpretation, IPCA factors reflect common sources of variation in managed portfolios, and the estimated factors are "portfolios of managed portfolios".

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Table of Contents

5 A New Norm Beyond PCA

- Recap
- Risk Premium PCA
- PCA with Economic Targets
- Instrumented PCA
- A New Norm Beyond PCA

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- Consider again the conditional latent factor model:

$$R_{i,t+1} = \boldsymbol{\beta}(\mathbf{C})_{i,t}^{\top} \boldsymbol{f}_{t+1} + e_{i,t+1},$$

  where $\mathbf{C}$ represent firm characteristics that are related to asset returns. In this framework, the work goes into modeling factor exposure $\boldsymbol{\beta}$.

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- From a **machine learning** perspective, $\boldsymbol{\beta}()$ maps a large information set into a small factor model ($\mathbb{R}^L \to \mathbb{R}^K, L \gg K$). Once we find $\boldsymbol{\beta}()$, we also find factor returns, i.e., $\boldsymbol{f}_{t+1} = (\boldsymbol{\beta}_t^{\top}\boldsymbol{\beta}_t)^{-1}\boldsymbol{\beta}_t^{\top}\boldsymbol{R}_{t+1}$.

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- Within this framework, the IPCA model can be viewed as linearly mapping firm characteristics to $\boldsymbol{\beta}$ via $\boldsymbol{\Gamma}$. However, it is evident that we need not limit ourselves to linear models.

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- Within this framework, the IPCA model can be viewed as linearly mapping firm characteristics to $\boldsymbol{\beta}$ via $\boldsymbol{\Gamma}$. However, it is evident that we need not limit ourselves to linear models.

- Factor model approach is equivalent to the SDF framework. Therefore, within a unified framework, we can analyze and compare various machine learning methods. (Next Lecture!!!)

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Questions?

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